

1. Let $\{X_n\}$ be a submartingale and T a stopping time such that $E[T] < \infty$. Assume

$$E[|X_{n+1} - X_n| | \mathcal{F}_n] \leq M \quad \text{on } (T > n)$$
 for some constant M . Show that X_T is integrable and $E[X_0] \leq E[X_T]$.
2. Let $X_n \rightarrow \underline{X}$ in probability. Let f be continuous on $\mathbb{R}^1$. Then $f(X_n) \rightarrow f(\underline{X})$ in probability. Prove this.
3. Can convergence in a.e. be given by a metric? In other words is there a metric ρ such that $\rho(X_n, 0) \rightarrow 0$ is equivalent to $X_n \rightarrow 0$ a.e. Substantiate.
4. Explain the notion of convergence in distribution. Let $X_n \rightarrow \underline{X}$ in distribution. Does it follow that $X_n \rightarrow \underline{X}$ in probability. Suppose $P[X = a] = 1$. Is it then true that $X_n \rightarrow \underline{X}$ in probability. Substantiate all your answers.
5. Let $\bar{Y} \in L^1$, $|X_n| \leq \bar{Y}$ and $X_n \rightarrow \underline{X}$ in distribution. Show that $E[|X_n|^p] \rightarrow E[|\underline{X}|^p]$.
6. Given that f is a characteristic function and $|f(\frac{1}{2})| = 1$ what can you conclude?

7. State and give a sketch of the proof of the Strong Law of Large Numbers.

8. Let $\bar{X}_n$ be i.i.d., $E[\bar{X}_n] = 0$, $E[\bar{X}_n^2] = 1$

Can this sequence be complete orthonormal.

Substantiate.

9. Let μ be a ^{non-trivial} infinitely divisible distribution on $\mathbb{R}^1$. Prove that μ cannot have compact support.